

Help Document

Wired technologies

The orders of the following wired technologies are, roughly, from slowest to fastest transmission speed.

Twisted pair wire is the most widely used medium for all telecommunication. Twisted-pair cabling consist of copper wires that are twisted into pairs. Ordinary telephone wires consist of two insulated copper wires twisted into pairs. Computer network cabling (wired Ethernet as defined by IEEE 802.3) consists of 4 pairs of copper cabling that can be utilized for both voice and data transmission. The use of two wires twisted together helps to reduce crosstalk and electromagnetic induction. The transmission speed ranges from 2 million bits per second to 10 billion bits per second. Twisted pair cabling comes in two forms: unshielded twisted pair (UTP) and shielded twisted-pair (STP). Each form comes in several category ratings, designed for use in various scenarios.

Coaxial cable is widely used for cable television systems, office buildings, and other work-sites for local area networks. The cables consist of copper or aluminum wire surrounded by an insulating layer (typically a flexible material with a high dielectric constant), which itself is surrounded by a conductive layer. The insulation helps minimize interference and distortion. Transmission speed ranges from 200 million bits per second to more than 500 million bits per second.

ITU-T G.hn technology uses existing home wiring (coaxial cable, phone lines and power lines) to create a high-speed (up to 1 Gigabit/s) local area network.

An optical fiber is a glass fiber. It uses pulses of light to transmit data. Some advantages of optical fibers over metal wires are less transmission loss, immunity from electromagnetic radiation, and very fast transmission speeds of up to trillions of bits per second. One can use different colors of lights to increase the number of messages being sent over a fiber optic cable.

Wireless technologies

Terrestrial microwave – Terrestrial microwave communication uses Earth-based transmitters and receivers resembling satellite dishes. Terrestrial microwaves are in the low-gigahertz range, which limits all communications to line-of-sight. Relay stations are spaced approximately 48 km (30 mi) apart.

Communications satellites – Satellites communicate via microwave radio waves, which are not deflected by the Earth's atmosphere. The satellites are stationed in space, typically in geosynchronous orbit 35,400 km (22,000 mi) above the equator. These Earth-orbiting systems are capable of receiving and relaying voice, data, and TV signals.

Cellular and PCS systems use several radio communications technologies. The systems divide the region covered into multiple geographic areas. Each area has a low-power transmitter or radio relay antenna device to relay calls from one area to the next area.

Radio and spread spectrum technologies – Wireless local area networks use a high-frequency radio technology similar to digital cellular and a low-frequency radio technology. Wireless LANs use spread spectrum technology to enable communication between multiple devices in a limited area. IEEE 802.11 defines a common flavor of open-standards wireless radio-wave technology known as Wifi.

Free-space optical communication uses visible or invisible light for communications. In most cases, line-of-sight propagation is used, which limits the physical positioning of communicating devices.

Network nodes

Apart from the physical communications media described above, networks comprise additional basic system building blocks, such as

1. network interface controller (NICs),
2. repeaters,
3. hubs,
4. bridges,
5. switches,

6. routers,
7. modems, and
8. firewalls.

Network interfaces

An ATM network interface in the form of an accessory card. Very many network interfaces are built-in.

A network interface controller (NIC) is computer hardware that provides a computer with the ability to access the transmission media, and has the ability to process low-level network information. For example the NIC may have a connector for accepting a cable, or an aerial for wireless transmission and reception, and the associated circuitry.

In addition, the NIC will have access to an identifier for either the NIC or the computer as a whole.

In Ethernet networks, each network interface controller has a unique Media Access Control (MAC) address which is usually stored in the controller's permanent memory. MAC address uniqueness is maintained and administered by the Institute of Electrical and Electronics Engineers (IEEE) in order to avoid address conflicts between devices on a network. The size of an Ethernet MAC address is six octets. The 3 most significant octets are reserved to identify NIC manufacturers. These manufacturers, using only their assigned prefixes, uniquely assign the 3 least-significant octets of every Ethernet interface they produce.

Repeaters and hubs

A repeater is an electronic device that receives a network signal, cleans it of unnecessary noise, and regenerates it. The signal is retransmitted at a higher power level, or to the other side of an obstruction, so that the signal can cover longer distances without degradation. In most twisted pair Ethernet configurations, repeaters are required for cable that runs longer than 100 meters. A repeater with multiple ports is known as a hub. Repeaters work on the physical layer of the OSI model. Repeaters require a small amount of time to regenerate the signal. This can cause a propagation delay which can affect network performance. As a result, many network architectures limit

the number of repeaters that can be used in a row, e.g., the Ethernet 5-4-3 rule.

Hubs have been mostly obsoleted by modern switches; but repeaters are used for long distance links, notably undersea cabling.

Bridges

A network bridge connects and filters traffic between two network segments at the data link layer (layer 2) of the OSI model to form a single network. This breaks the network's collision domain but maintains a unified broadcast domain. Network segmentation breaks down a large, congested network into an aggregation of smaller, more efficient networks.

Bridges come in three basic types:

Local bridges: Directly connect LANs

Remote bridges: Can be used to create a wide area network (WAN) link between LANs. Remote bridges, where the connecting link is slower than the end networks, largely have been replaced with routers.

Wireless bridges: Can be used to join LANs or connect remote devices to LANs.

Switches

A network switch is a device that forwards and filters OSI layer 2 datagrams between ports based on the MAC addresses in the packets.[8] A switch is distinct from a hub in that it only forwards the frames to the physical ports involved in the communication rather than all ports connected. It can be thought of as a multi-port bridge.[9] It learns to associate physical ports to MAC addresses by examining the source addresses of received frames. If an unknown destination is targeted, the switch broadcasts to all ports but the source. Switches normally have numerous ports, facilitating a star topology for devices, and cascading additional switches.

Multi-layer switches are capable of routing based on layer 3 addressing or additional logical levels. The term switch is often used loosely to include devices such as routers and bridges, as well as devices that may distribute traffic based on load or based on application content (e.g., a Web URL identifier).

Routers

A typical home or small office router showing the ADSL telephone line and Ethernet network cable connections

A router is an internetworking device that forwards packets between networks by processing the routing information included in the packet or datagram (Internet protocol information from layer 3). The routing information is often processed in conjunction with the routing table (or forwarding table). A router uses its routing table to determine where to forward packets. (A destination in a routing table can include a "null" interface, also known as the "black hole" interface because data can go into it, however, no further processing is done for said data.)

Modems

Modems (MODulator-DEModulator) are used to connect network nodes via wire not originally designed for digital network traffic, or for wireless. To do this one or more frequencies are modulated by the digital signal to produce an analog signal that can be tailored to give the required properties for transmission. Modems are commonly used for telephone lines, using a Digital Subscriber Line technology.

Firewalls

A firewall is a network device for controlling network security and access rules. Firewalls are typically configured to reject access requests from unrecognized sources while allowing actions from recognized ones. The vital role firewalls play in network security grows in parallel with the constant increase in cyber attacks.

Networking Commands

1. arp

Displays the 'Address Resolution Protocol' table e.g.

arp -a

displays all arp entries for all connected devices.

`arp -d <ip address>`

deletes the arp entry for that particular IP address.

2. ftp

The command **file transfer protocol** attaches you to another IP device e.g.

`ftp 141.205.15.154`

attaches you to the device with address 141.205.15.154. You are normally presented with a login and password screen.

Commands that are used in FTP are:

- **dir** - directory listing.
- **quit** - quit from ftp.
- **cd** - change directory.
- **get** or **mget** - get a file (or multiple files).
- **put** or **mput** - put a file (or multiple files).
- **bin** - sets up your system to receive binary files.
- **hash** - displays hashes whilst files are being transferred.
- **lcd** - local change directory changes the directory on your local machine to which you are sending and receiving files. This is useful as it saves you having to quit ftp to carry out the directory change.

The **Hosts** file can be found in the directory '/etc'.

3. netstat

This stands for **network statistics**, e.g.

`netstat -r`

displays the routing table of the Unix box.

`netstat -a`

displays all network information.

Unix uses **routed** to listen to RIP in order to discover the Default Gateway.

4. ping

Ping an IP device e.g.

`ping 141.205.51.26`

5. rlogin

This works like telnet, e.g.

`rlogin 141.205.52.16`

takes you to another Unix machine only. To quit you press 'return', '~', '.' and 'return' again.

6. telnet

Ctrl-6 and then **Ctrl-]** gets you to the **telnet>** prompt where typing **close** gets you out of telnet.

7. ipconfig

This displays the IP configuration of the box, e.g.

`ipconfig -a`

displays all IP configuration.

If you want to look at the routing process you can type:

`ps -ef | grep routed`

to send the 'routed' information to a file.

8. More Network Commands on Linux / Unix

1. ftp : Transfer files to another host (insecure)
 2. host: Get information about networked host.
 3. ipconfig : Display Ip address information.
 4. netstat: Display routing information and network statistics
 5. ip: Display Ip address information.
 6. ping: Sends answer request to other hosts.
 7. rdesktop: Display MS Windows desktop on Linux Machine
 8. route: Show routing information
 9. scp: Secure copy files to and from another host.
 - 10.sftp: Secured ftp
 - 11.ssh: Make an encrypted connection to another host.
 - 12.ssh-keygen: Generates authentication keys for SSH
 - 13.telnet: Make an insecure connection to another host.
 - 14.tracepath/traceroute: Prints the packet and time taken for hops to a host.
 - 15.whois: Get information about a domain name.
 - 16.xclock: X window clock application, handy for testing remote display.
 - 17.xhost: X window access control tool.
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